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Model selection with considering the CO_2 emission alone the global supply chain

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Abstract This study formulates a model for analyzing eco-environmental impact on global supply chain network. The multi-criteria optimization model is applied to seek optimal solutions that not only can achieve predetermined objectives, but also can satisfy constraints for multi-product problems. The overall optimization is achieved using mathematical programming for modeling the supply chain functions such as location, inventory, production, distribution functions and transportation mode selections. Then, the supply chain model is formulated as a minimization problem for costs and environmental impacts. Herein, the solution is the flow of goods in global supply chain environment in different periods of time over one year. Furthermore, the numerical values obtained from a real company are applied to these mathematical formulations to test its usability. The testing is conducted in four different cases that include two combinations, no due date constraint and due date constraint, without connection of distributor and with connection of distributors. The results from these experiments can help in determining the best transportation routes, inventory levels, shipment quantity, and transportation modes. Specifically, the results propose a new configuration for designing global supply chain for the case company that could minimize economical and environmental impacts problems simultaneously.

Keywords Global supply chain · Supply chain management · Environmental management · CO_2 emission · Multi-criteria optimization model · Integer linear programming

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Introduction

Global warming, greenhouse effect, high sea level, pollutions, ozone layer depletion and natural resources exhaustion are the major environmental issues and concerns all around the world. As a result, consumers are putting more consideration on product information, and even on the companies producing them to make sure the products they buy have less impact on the environment. Specifically, they demand environment-friendly products with clean production process. Therefore, a modern company has adjusted and reformed its business operations and production process to manage it sustainably. Furthermore, the adoption of new regulations (Carbon tax) in some of the European countries has created a challenge for managers to reduce CO_2 emissions in daily business operation. According to International Energy Agency report (2009), the annual carbon dioxide emission from transport activities accounts for approximately 30% global share.

From the above analysis, this study attempts to design a supply chain model that could minimize economical and environmental impacts problems simultaneously. The objective of this study is to identify and analyze the flow of product from its raw materials to the end-consumer throughout the logistical activities such as freight transport, storage, inventory management, materials handling and selection of suppliers to create an optimal supply chain model that consider the economic and environmental issues at the same time. This is done by using an optimization model, an integer linear programming that connects these issues together in objective function. In addition, the production condition, demand, lead

time, suppliers, and company capacity issues are all covered by the constraint functions. In this context, a Taiwan hand-tool company is chosen to investigate the cost flow and CO_2 emission that are results of their daily business operation. Herein, the distribution system of the company with locations in different countries such as India, France, Germany, USA, etc., created an attractive case for our study. These distributors in different countries can supply and receive products to each other. Finally, the comparison between the current and proposed configuration of the company's supply chain are conducted to test and prove the feasibility of our model.

The paper is organized as follow. In section 2 we set out a theoretical review of studies that are relative to Supply Chain Management (SCM), Global SCM, and environmental management in SCM. In section "Problem definition", we focus on problem definition. In section "Empirical case", we present our multi-criteria supply chain optimization model. The application of the model for a problem example is presented in section "The model". Finally, in section "Result", we present our main conclusions.

Literature review

Global supply chain network

Although SCM and supply chain network design are not new topics, they still receive considerable interest among the supply practitioners and academicians. This is partly due to the fact that an effective SCM and supply chain network design not only could bring essential benefits expected by most companies (such as reduced cost and improved cycle time), but also can help company to increase competitive advantage dynamic (Viswanadham and Gaonkar 2003).

SCM is a process which "encompasses every effort involving producing and delivering a final product or service, from the supplier's supplier to the customer's customer. SCM includes managing supply and demand, sourcing raw materials and parts, manufacturing and assembly, warehousing and inventory tracking, order entry and order management, distribution across all channels, and delivery to the customer" (Nikbakhsh 2009). Supply chain design is the process of determining the supply chain infrastructure including the plants, distribution centers, transportation modes and lanes, production processes, etc. that will be used to satisfy customer demands (Harrison 2001).

Literature is rich in contributions referring to the supply chain networks. Jang et al. (2002) proposed a system for supply network management that can determine the production and inventory quantity, the transportation quantity, and the choice of plants that should be operated. Beamon and Fernandes (2004) designed a supply chain recovery network to determine which warehouses and collection centers should be operated, to identify capacities of storage and how much material should be transported between each pair of sites in order to minimize the present loss. Chan and Chan (2005) used simulation model for designing supply chain. They built and tested five different supply chain models with the aim to determine the supply chain models that could achieve the optimal performance, in terms of inventory level, order lead time, resources utilization, and transportation costs. Bottani and Montanari (2010) used a discrete-event simulation model in order to examine the effects of different supply chain configurations in term of bullwhip effect, holding cost, stock-out cost, order cost, transport cost and shipping/receiving cost.

Extending these concepts, global supply chainsare "organization networks that are involved, through upstream and downstream linkages, in different processes and activities that produce value in the form of products and services to the ultimate customer" (Magnusson et al. 2008). Prasad and Sounderpandian (2003) examined the literature on international operation management in order to identify factors that influence the performance of global supply chain. The authors investigated and organized the factors in three categories (by country, industry and company) to work throughout the supply chain areas such as procurement, processing and distribution to achieve the competitive advantages in cost, quality, lead time and innovation issues. This research is our guideline of how to determine the factors that we should consider in global SCM. Meixell and Gargeya (2005) conducted a literature review about global supply chain design in 5 years periods for the models developed prior to 1990, in 1990–1995 period, in the 1996–2000 periods, and after 2000. The results showed that most of models have used location, material flows, supplier selection, product allocation, production shifting, and transport mode as decision variables. Furthermore, profit, operating profits or after-tax profit, corporate taxes and transfer prices, minimize cost, and minimize time are considered as the key performance factors or objective function in these models. More recently, Gumus et al. (2009) investigated and developed a global supply chain network that combined artificial neural network (ANN) simulations for demand forecast and mixed integer linear programming for calculating the optimal product flow between the factories, warehouses and distributors. The solution of their model is decision making process to open or close factories and warehouses and to define minimum cost production quantity for supply products from factories to warehouses and from warehouses to distributors. However, transportation cost and transport mode were not considered in their paper. One of the most comprehensive models at present is the research of Reiter et al. (2010). The authors proposed a framework that integrates transport information into production planning and control systems along with the global supply chain. The objective function minimizes the costs incurred due to delaying order delivery to customers, processing and storage

costs of orders and fixed and variable costs of each selected routing. Their study pointed out the potential of integrating production scheduling and logistics activities. However, because of the large amount of data that needs to be considered and the number of decisions, only a small test scenario of a global supply chain which includes one production facility and a transportation network within Germany was obtained.

Generally, the numbers, locations, and capacities of manufacturing, assembly, and distribution facilities and then the flow of the materials from selected suppliers to customers, including inventory levels and transportation modes, and then international trade rules and financial issues are criteria factors that are considered in the global SCM. Moreover, the integrating production scheduling and logistics activities are an attractive topic to investigate for the global supply chain network.

Environmental management in supply chain management

Climate change is one of most important global issues nowadays. Environmental management can be defined as the management of human's interactions with environment and their impacts on environment (Nikbakhsh 2009). Therefore, environmental management in supply chain is the management approach that defines how a supply chain can address its impact on the natural environment. A common challenge for environmental management is the conviction of the manager it will increase company's costs. Walton et al. (1998) has pointed out that companies that want to get greater benefits from their environmental management processes must integrate other members of the supply chain into these processes.

Handfield et al. (2005) demonstrated the alternative way for companies to begin the integration of environmental and supply chain decisions. With a plenty of practical cases such as Sony, Samsung, Intel, Haworth, Doosan, etc., and an in-depth analysis on how these companies perform environmental management, their paper provided guidelines for managers for managing the integration of environmental issues and supply chain. Yang et al. (2009) designed the closed-loop optimal supply chain network which includes raw material suppliers, manufacturers, retailers, consumers and recovery centers, with consideration of both upstream and downstream side of supply chain network. The objective of this model was to find the supply chain network that aims at the profit maximization. Bojarski et al. (2009) presented an approach for supply chain design focusing on environmental impacts and their sources. The optimization model was conducted to select the appropriate technology, or the appropriate raw material/service suppliers. In their research, we calculate the environmental impacts indicators of supply chain network (e.g. direct emissions, purchased energy emissions, raw materials production emissions and transport distribution emissions).

Munksgaard (2001) incestigated the CO_2 emission responsibility of producer or the consumer in international trade, and compare two accounting models which is the production account model and the consumption account model. From other perspective, Lee et al. (2008) studied the impact of carbon tax in Taiwan, and found that too low carbon rate would reduce the effectiveness of CO_2 reduction, while too high rate may result in serious impact on the industry's production (Hua 2000; Siriwardena 2007; Lee 2008). Olsthoorn (2001) investigated the CO_2 emission on future transport pattern by targeting on CO_2 emission of different transport mode.

On the other hand, controlling carbon emissions and measuring carbon footprint across a supply chain is also a challenge for organizations (Sundarakani et al. 2010). The objective function of this model is to minimize the total cost, energy, CO_2 emission and cycle time. A carbon footprint is calculated in transportation process between each stage of supply chain. Since the main focus of this model is network configuration, therefore, scheduling, quantity demand, and flow of material and goods does not receive much attention. Then, Piecyk and McKinnon (2010) built a model to forecast a series of baseline trends in logistics and SCM and associated environmental effects up to year 2020 through examining the CO_2 emission on road freight transport. The paper pointed out that there is a need for companies to understand the measurement method and management of the CO_2 emissions of their road fleets. Sundarakani et al. (2010) considered the total carbon emission across the supply chain should consist of stationary sources (emissions from material processing, manufacturing, and warehousing) and non-stationary sources (emissions from inbound and outbound logistics). In the end, the authors recommended that the companies consider following issues: (i) design products and supply chains concurrently with carbon emissions in mind; (ii) add carbon emission rates to supplier selection criteria; (iii) have green supply and purchasing policies; (iv) maintain acceptable carbon regulation at the manufacturing level; (v) leverage innovation in logistics services to reduce carbon emissions; (vi) reduce inventory and increase visibility at the distribution level; (vii) have green packaging and distribution strategies; (viii) reduce, reuse and recycle at the consumption stage; (ix) create awareness among consumers on carbon emissions.

Luo et al. (2001) is the one of pioneers that used a mathematical model for designing and optimization of supply chains in terms of product cost, cycle time, quality, energy and environmental impact with the focus on SCM. The environmental factors mentioned in this paper are energy consumption and CO_2 emissions. Following this idea, Dotoli et al. (2006) proposed a single- and multi-objective optimization model to solve this problem. Comparing the results obtained by solving fuzzy multi-objective optimization method, the authors found that the model using single-objective can guarantee the optimum solutions.

Cholette and Venkat (2009) used a web-based tool to calculate the energy and carbon emissions associated with each transportation links and warehousing activities in the wine industry. They concluded that the energy consumption and carbon emission can vary substantially, depending on the supply chain configuration. Thus, the managers can select a suitable design depending on the current supply chain scenario.

This research attempts to design a supply chain network that not only considers the important factors of a traditional supply chain (i.e. economic dimension including time, cost, scheduling, routing and transport mode selection), but also deals with environmental responsible issue, such as CO_2 emissions.

Problem definition

Description of problem

In order to formulate a global supply chain network, a comprehensive conceptual framework which involves raw material suppliers, component suppliers, focal company, distribution center, and customer is proposed as shown in Fig. 1. The term "global" used here implies that members of the supply chain are located in different countries.

In Fig. 1, the company in the middle is referred to as the focal company that is an end-assembly company; it can be any of the firms involved in the supply chain, depending on the frame of reference of the manager viewing the diagram. However, when we mention the party that plays the role of controlling the activities in the supply chain, we refer to it as "the case company" (note that in this study case, the case company and the focal company are the same organization). Besides, the discontinuous line in the model represents the order of the focal company to the raw material suppliers. These raw materials suppliers do not deliver materials to the case company, they supply to the component supplier of the case company instead, e.g. the focal company sends order to the raw material supplier, then this raw material supplier will supply to the component suppliers (from $S_{n,1}$ and $S_{n,m}$). The continuous line in the model represents for the order of the members in downstream (e.g. focal company, distributors, customers) to members in upstream (e.g. suppliers, the case company) and then the members in upstream supplies goods directly to members in downstream. For example, the continuous line from the component supplier S_{11} to the focal company means that the focal company will send orders to the component supplier S_{11}, and after order lead time, this component supplier will supply component to the focal company.

Generally, the supply chain members include raw material suppliers, component suppliers, focal company, distribution center and customers which are coded as S_i, S_{ij}, the focal company, D_d and C_r, respectively. The raw material suppli-

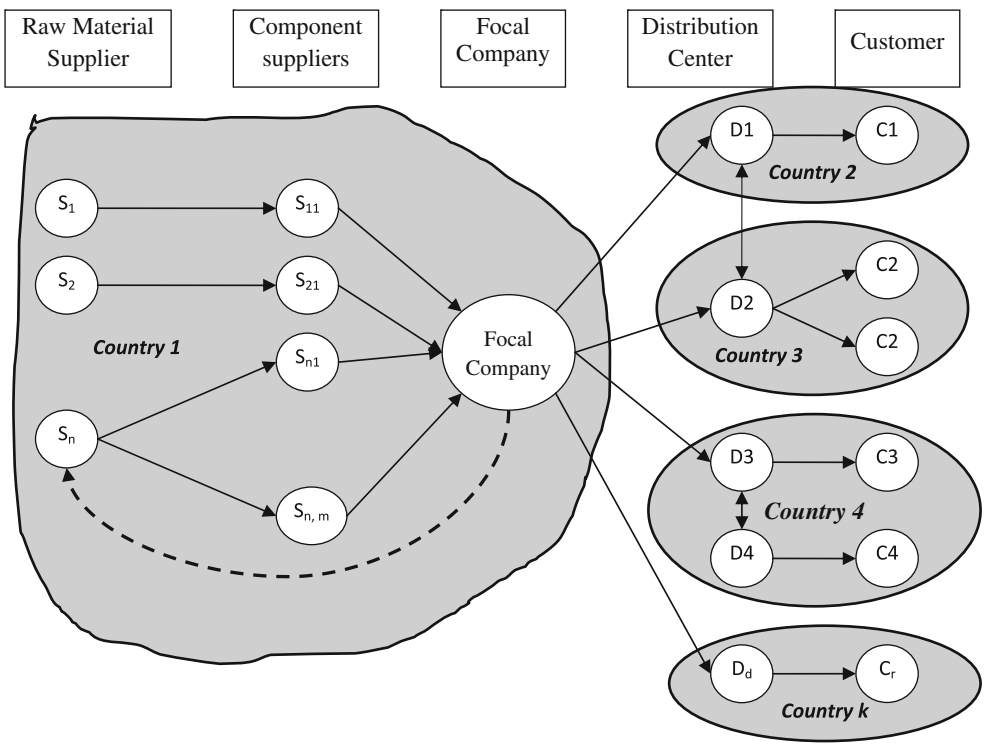


Fig. 1 The global supply chain network in our research

ers (S_i) will deliver material to component material supplier (S_{ij}). These raw material suppliers can receive orders from company displayed on discontinuous line, or from component suppliers. In addition, the flow of goods is represented by continuous lines. After receiving materials, the component material suppliers will produce components based on focal company's design and volume. Based on the preferred lead time, these members will deliver to focal company. The focal company will check these components, assemble them together, then put their own brand or customer brand name on it, and finally pack it. According to orders, the volume and due date, the focal company will ship its products to distributors who are located in different countries (D_d) and who then sell to customers (C_r).

Furthermore, all the case company's components are outsourced, the production activities of the focal company is only limited to quality control and packaging. This paper does not look into the production process, but rather the flow of goods, from suppliers to customers, is our target. Instead of focusing on production cost, the component purchasing price will be considered.

To specify the study scope and facilitate model formulation, assumptions are postulated as below.

(i) We only consider CO_2 emissions incurred by transport modes;
(ii) The time horizon for the model is divided in 12, 1 year, periods;
(iii) The air mode was assumed to be three times as expensive, but from four to eight times faster than the sea mode;
(iv) We ignore the difference in duties between each continent;
(v) Initial inventory is zero;
(vi) Ratio of materials and products is 1:1;
(vii) Facility capacity such as the production capacity and supplier capacity, lead time, annual demand is given;
(viii) All transportation time is both positive and deterministic; the whole transportation process is supposed to take place within the same period, in other words, transportation time is less than the duration of a period;
(ix) The values of Ibs CO_2 emission per ton-mile using truck, ship, or air are 0.3725, 0.0887, and 1.7739 for each mode;
(x) No matter what country the products are shipped to, the tax and duty are not considered in our model;
(xi) The focal company buys materials from domestic suppliers, thus the focal company and suppliers are from the same country; then, the focal company exports its products to several countries in Asia, Europe, and USA; the distributors located in different countries can then supply the products to other distributors or customers;
(xii) To gain local product advantage, the focal company purchases materials from domestic suppliers; furthermore, we assume that the distance between focal company and suppliers is close; so the transportation cost between suppliers and focal company is close to zero.
(xiii) Considering about the Emission Trading, we propose CO_2 may be priced as Carbon Trading.

The challenge for this global supply chain network is selection of raw material and component suppliers, transportation modes, distributor facilities and the flow of goods in order to meet the due delivery dates set by customers, minimize the cost of network and the quantity of emitted CO_2. Moreover, due to the capacity constraints in the supply chain decisions, production and company planning activities need to meet the lead time of customers. The potential solutions are either storing products in warehouse that charges a holding fee or using the faster transportation mode which increases transportation fee. There can be significant cost savings through proposed by our model, especially in terms of synchronization of activities leading to reduced inventory levels or CO_2 emission or transportation cost.

In next section, we introduce and analyze the European market and supply chain, since this is where the case company supply chain is mostly located.

European scenario

For the case company, the majority of its customers are located in Europe. The European Union (EU) composes of 27 independent members states that are managed by European Communities. The purpose of the EU is to enhance political, economic and social co-operation (Europa Commission 2010). Therefore, one of the advantages of the EU member states is the " Single Market" rule that ensure the free movement of people, goods, services and capital adopted with aiming to "bring down barriers and simplify existing rules" in the whole EU (Europa Commission 2010). In other words, there are no limits on what you can buy and take with while crossing the borders within the EU countries. If the goods originate from countries inside the EU, no duty is payable. In addition, if the goods originate from countries outside the EU, duty is only counted one time no matter what how many other the EU countries they can be pass (Business Link website 2010). And also, a fixed duty is applied for all of the EU countries (European Commission 2010d; Encyclopedia of the Nations 2010). In sum, trading between EU countries has following advantages: fixed duty, one time payable duty, ease of moving, elimination of the exchange currency rate problems.

Empirical case

Company information

To solve the multi-product models we have used Lingo 11 software. Furthermore, a case study with real data from a Taiwanese hand-tool company, Chewren, was used to demonstrate the applicability of this model. This Taiwanese company undertakes two positions in our model. First, it is an end-assembly company that conducts components testing, component assembly to final product and packaging, and we refer to it as "the focal company". The other position is the head-quarter company that controls the entire operation of the supply chain; we then refer to it as "the case company".

Our case consists of one focal company, nine distributors, eleven customers and two products, as shown in Fig. 2. Herein, the suppliers and the focal company are located nearby, which means reduced lead time, transportation cost, import tax, etc. These nine independent distributors are treated as the focal company's customers that are located in different countries, namely France, Germany, Italy, Sweden, Austria, USA, India and Japan. Depending on the demand of each country and geographical distance, one or two distributors are set up, and products are delivered to customers from these independent distributors. Decisions that should be made are the configuration of supply chain network for each period, the flow of goods that this network can carry, and the transportation modes used to deliver goods to each node and the inventory quantity at each stage (the focal company,

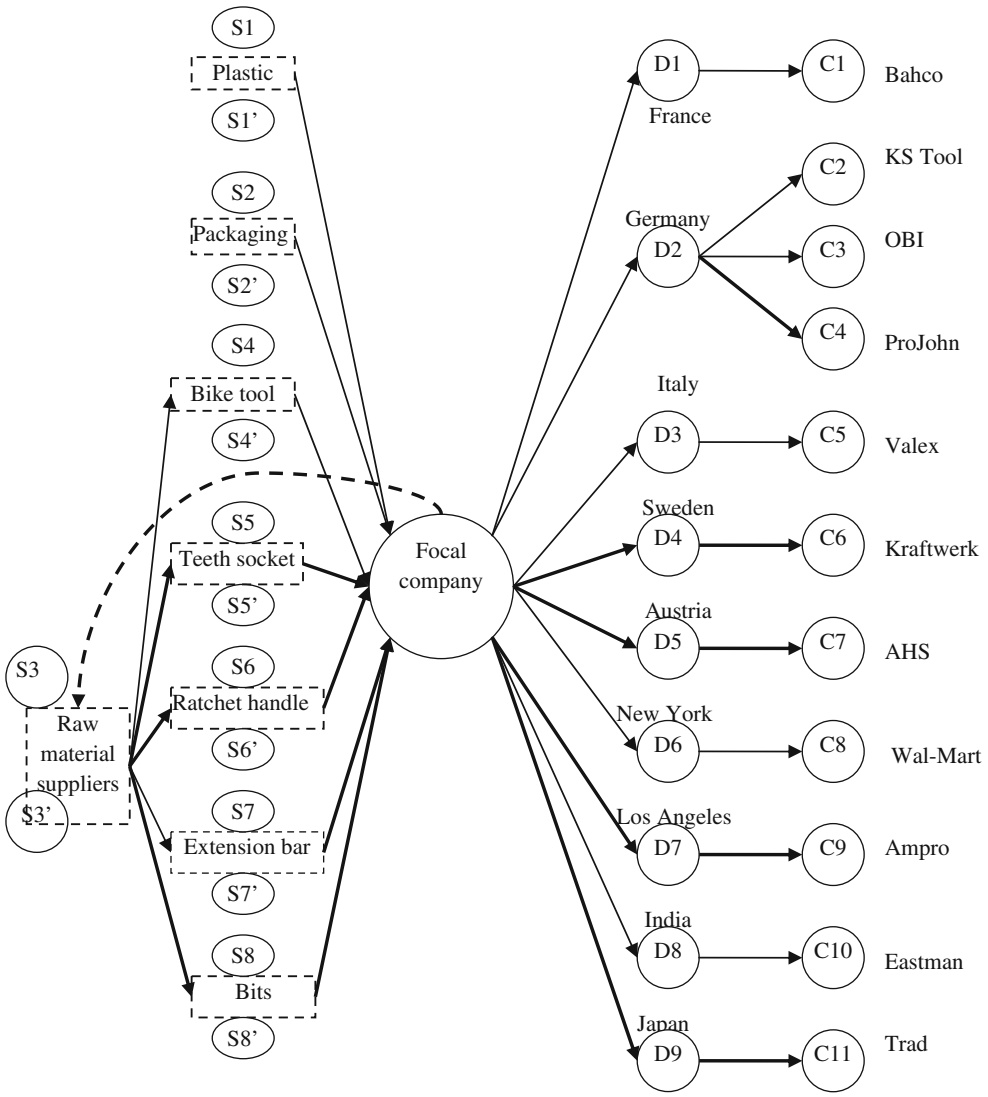


Fig. 2 Global supply chain network of the case company: two products, without link between distributors

distributors). These decisions are controlled by an objective function that minimizes the total cost and CO_2 emission. But the total cost and the CO_2 emission are two independent criteria, one (cost) is calculated American Dollars (USD), and the other (CO_2 emission) in kilograms (kg). The scalar-valued weights (λ) of multi-criteria global optimization are used to adjust to the objective function so that it matches the decision maker's preferences. This approach is used to deal with independent criteria issue (Kim and de Weck 2005; Belgasmi et al. 2008). We call $\lambda 1$ the scalar-valued weights of cost, and $\lambda 2$ the scalar-valued weights of CO_2 emission. The sum of scalar-valued weights (λ) is 1 (Kim and de Weck 2005), thus $\lambda 1 + \lambda 2 = 1$. Within the objective function, we assume that the decision maker's preference between cost reduction and CO_2 emission are 3:1, namely, $\lambda 1$ is 0.75 or 75% and, $\lambda 2$ is 0.25 or 25%. In other words, the decision maker wants to seek an optimal solution that can achieve 75% cost reductions and 25% CO_2 emission reductions. However, the λ value is a bias, and it can be changed to meet market changes, government pressure, and customer requirements, therefore it is not a fixed value. The value we applied in this study is based on the opinion of manager who is in charge of decision making of the case company.

Product information

Product 1: "17 in 1 Nylon Folding Bike Tool Set"

"17 in 1 Nylon Folding Bike Tool Set" is one of the case company's best-selling products. The finished product includes two main component types: plastic parts and metal parts. The case company buys metal from the metal company ($S3$, $S3'$ in Fig. 2) and then this supplier delivers directly to other suppliers (OEM companies $S4$, $S4'$ in Fig. 2) in order to produce five sub-components which include hex keys, open wrenches, spock wrenches and screwdrivers. In addition, to be effective in reducing inventory carrying costs and production costs, the plastic parts and the packages are also outsourced by the plastic suppliers and the packaging suppliers ($S1$, $S1'$ and $S2$, $S2'$ in Fig. 2). This product is aimed for bicyclists for repair of their bicycles while on the road. Because of convenience, unique design and suitable price, this product is sold around the world in large amounts. Therefore, finding a solution to satisfy all demand at different periods, gain minimum cost and reducing environmental impacts are important issues for the company.

Product 2: "28PC Bit Ratchet & Socket Sets"

The second product used for our model is a set of reversible ratchet bit drivers, bit holders, permanent magnets, hex machine fittings, bits adapters and 29 1/4"-bits. All of them are put in a plastic case. This product includes three main components: metal parts, plastic case and packaging. As we mentioned before, the case company takes advantage of outsourcing activities: all of them are purchased from OEM companies. The case company sends designs and order volumes to these OEM companies, and after 15–30 days, these parts are supplied to the focal company. Herein, the metal parts, which are the main component of this product, will be conducted as follows. Firstly, the case company sends order to the metal company to purchase the metal. Within 1–5 days, the metal company delivers the metal, such as teeth sockets, ratchet handles, extension bars and bits, to the OEM factory to produce these components respectively. After 15–25 days, these factories supply these components to the focal company. Then, the focal company checks and assembles them and sends final products to its distributors and customers.

Product routing

At the beginning of each period, the case company conducts production planning. For example, how much material quantity is available in the warehouse, how many products are in stock, what is demand of this period, etc. Based on this information, the case company sends orders to its suppliers. The orders for plastic components and packaging of product 1 and product 2 are sent to the plastic suppliers and packaging suppliers respectively. In other words, product 1 and 2 has the same suppliers for plastic and packaging ($S1$ or $S1'$ and $S2$ or $S2'$ in Fig. 2). On the other hand, the case company sends its orders to metal company ($S3$ or $S3'$) to supply metal to factories that produce the teeth sockets ($S5$ or $S5'$), ratchet handles ($S6$ or $S6'$), extension bars ($S7$ or $S7'$), and bits ($S8$ or $S8'$) components for product 2, and the bike tool factory that produces the hex keys, wrenches, and screw drivers ($S4$ or $S4'$) for product 1. Simultaneously, these factories also receive orders from the case company to produce components within the time that the case company desires. From 15 to 30 days, the focal company receives all components and material from these suppliers. Depending on the demand at each period, they are stocked in the warehouse or processed immediately.

Besides sending orders to suppliers and handling production activities, the case company needs to fulfill the orders in each period. Based on data collected from the case company, product 1 is delivered to distributor 1 (France), distributor 2 (Germany), distributor 3 (Italy), distributor 6 (New York) and distributor 8 (India). From these distributors, Product 1 is supplied to customer 1 (Bahco), customer 2 (KS Tool), customer 3 (OBI), customer 5 (Valex), customer 8 (Wal-Mart) and customer 10 (Eastman). Product 2 is delivered to distributors 4 (Sweden), distributor 5 (Austria),

distributor 7 (Los Angeles) and distributor 9 (Japan), then supplied to customer 4 (ProJohn), customer 6 (Kraftwerk), customer 7 (AHS), customer 9 (Ampro) and customer 11 (Trad).

The case company

The data obtained from the case company included monthly demand for over 1 year for six customers, transportation cost of supplying one product from the focal company to the distributors, and from the distributors to the customers, storage costs of one product, material cost for each type of materials, materials' order lead times, production time, and production capacity, as shown in Tables 1, 2, 3 and 4.

The following part presents the implementation of the model in two situations. The first one is that we seek the optimal solution in the current model of the case company that is without connection among distributors. The other one is that we propose a new supply chain configuration for the case company that has connections among distributors. Furthermore, to match with the real life, we split each situation into two sub-cases, one with no due date constraint and the other with the due date constraint. In sum, all four cases are presented in Table 5 below.

Table 1 Supplier data of the case company's global supply chain: two-product model

Material code	Material name	Supplier	Price (USD/unit)		Order lead time (days)
			Product 1	Product 2	
M1	Plastic	S1	0.048	1.276	15
		S1'	0.045	1.213	30
M2	Packaging	S2	0.048	0.702	15
		S2'	0.045	0.638	30
M3	Metal	S3	0.23	0.16	1
		S3'	0.223	0.144	5
M4	Bike tool	S4	1.436	NA	15
		S4'	1.364	NA	25
M5	Teeth socket	S5	NA	1.914	15
		S5'	NA	1.851	25
M6	Ratchet handle	S6	NA	3.191	15
		S6'	NA	3.031	25
M7	Extension bar	S7	NA	0.255	15
		S7'	NA	0.239	25
M8	Bits	S8	NA	0.479	15
		S8'	NA	0.447	25

Table 2 Monthly demand of the case company of product 1 over 12 periods (1 year)

Period*	Cus1**	Cus2	Cus3	Cus4	Cus5	Cus6	Cus7	Cus8	Cus9	Cus10	Cus11
1	0	0	0	0	0	0	0	0	0	0	0
2	0	100, 000	6,500	0	0	0	0	16, 000	0	16, 000	0
3	10, 000	100, 000	6,500	0	0	0	0	18, 000	0	0	0
4	0	100, 000	6,500	0	50, 000	0	0	16, 000	0	16, 000	0
5	0	100, 000	6,500	0	0	0	0	16, 000	0	0	0
6	10, 000	100, 000	8,000	0	0	0	0	18, 000	0	18, 000	0
7	0	100, 000	6,500	0	0	0	0	16, 000	0	0	0
8	0	100, 000	6,500	0	50, 000	0	0	16, 000	0	16, 000	0
9	10, 000	100, 000	6,500	0	0	0	0	18, 000	0	0	0
10	0	100, 000	6,500	0	0	0	0	16, 000	0	16, 000	0
11	0	100, 000	6,500	0	0	0	0	16, 000	0	0	0
12	10, 000	200, 000	13,500	0	50, 000	0	0	34, 000	0	18, 000	0

Table 3 Monthly demand of the case company of product 2 over 12 periods (1 year)

Period*	Cus1**	Cus2	Cus3	Cus4	Cus5	Cus6	Cus7	Cus8	Cus9	Cus10	Cus11
1	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	1,000	0	2,500
3	0	0	0	3,500	0	0	0	0	0	0	2,500
4	0	0	0	0	0	6,000	0	0	1,000	0	2,500
5	0	0	0	0	0	0	0	0	0	0	2,500
6	0	0	0	4,000	0	0	5,000	0	1,000	0	2,500
7	0	0	0	0	0	0	0	0	0	0	2,500
8	0	0	0	0	0	7,000	0	0	1,000	0	2,500
9	0	0	0	3,500	0	0	0	0	0	0	2,500
10	0	0	0	0	0	0	0	0	1,000	0	2,500
11	0	0	0	0	0	0	0	0	0	0	2,500
12	0	0	0	4,000	0	7,000	5,000	0	1,000	0	5,000

* There are twelve periods (from 1 to 12) that are studied in the case company
** There are eleven customers called Cus1–Cus11

Table 4 Parameter value of global supply chain network: two-product model

No	Parameter	Description	*Value*
1	P	Total product types	2
2	$Comp_1$	Total components of product 1	3
3	$Comp_2$	Total components of product 2	6
4	S	Total supplier quantity	16
5	D	The number of distributors	9
6	C	The number of customers	11
7	V	Transport modes that is used to deliver products from the focal company to distributor (ship, air)	2
8	V'	Transport modes that is used to deliver products from distributor to distributor (truck, ship, air)	3
9	T	The number of periods	12
10	D	The number of days in one period	30
12	Q_p	The minimum number of product p that can be shipped.	500

Table 5 The studied situations in multiple product model

Case	Due date constraint	Distributor connection
1	No	No
2	Yes	No
3	No	Yes
4	Yes	Yes

The model

We develop an integer linear programming model for the analysis of eco-environmental impact on a global supply chain network. We assume that the information is shared on the entire global supply chain in order to facilitate selecting partners and synchronize the material flowing through the network efficiently.

For the parameters description for the multiple product model, please refer to Appendix A and for descriptions of the used variables, to Appendix B.

Objective function definition

The whole incurred cost and quantity of CO_2 emission in the global supply network are calculated as follows.

i. *Total Material Cost*

$$\begin{aligned}\mathrm{MaC} = &\sum_{t=1}^{T}\sum_{p=1}^{P}\sum_{c=1}^{Comp_p}\sum_{m=1}^{Raw_{p,c}}\sum_{j=1}^{S_{p,c}}\sum_{i=1}^{S'_{p,m,c,j}} \mathrm{Pr}'_{p,m,c,i} \\ &* q_Raw_{t,p,m,c,i,j} \\ &+\sum_{t=1}^{T}\sum_{p=1}^{P}\sum_{c=1}^{Comp_p}\sum_{j=1}^{S_{p,c}} \mathrm{Pr}_{p,c,j} * q_Comp_{t,p,c,j} \qquad (1)\end{aligned}$$

ii. *Inventory Cost*

$$\begin{aligned}\text{InvC} &= \sum_{t=1}^{T}\sum_{p=1}^{P}\sum_{c=1}^{Comp_P} h_{p,c}\,.\,Inv_M_{t,p,c} \\ &+ \sum_{t=1}^{T}\sum_{p=1}^{P} h_p\, Inv_C_{t,p} \\ &+ \sum_{t=1}^{T}\sum_{p=1}^{P}\sum_{d=1}^{D} h_{p,d}\, Inv_D_{t,p,d} \end{aligned} \quad (2)$$

iii. *Total Transportation Cost*

In the multi-product model, total transportation cost is equal to the transportation cost per product p multiplied by number of product p shipped, and then plus fixed transportation cost. Therefore, if product $p1$ and $p2$ are shipped at the same period t, we only count it once.

$$\begin{aligned} TTrC = &\left(\sum_{t=1}^{T}\sum_{p=1}^{P}\sum_{v=1}^{V}\sum_{d=1}^{D} q_PD_{t,p,v,d} * TrC_PD_{p,v,d} \right. \\ &\left. + \sum_{t=1}^{T}\sum_{v=1}^{V}\sum_{d=1}^{D} OZ_PD_{t,v,d} * TrFC_PD_{v,d}\right) \\ &+ \left(\sum_{t=1}^{T}\sum_{v=1}^{V}\sum_{d=1}^{D}\sum_{k=1}^{D1_d} q_DD_{t,p,v,d,k} \right. \\ &* TrC_DD_{p,v,d,k} + \sum_{t=1}^{T}\sum_{v=1}^{V}\sum_{d=1}^{D}\sum_{k=1}^{D1_d} OZ_DD_{t,v,d,k} \\ &\left. * TrFC_DD_{v,d,k}\right) \\ &+ \left(\sum_{t=1}^{12}\sum_{p=1}^{P}\sum_{d=1}^{D}\sum_{r=1}^{R_d} q_DC_{t,p,d,r} * TrC_DC_{p,d,r} \right. \\ &\left. + \sum_{t=1}^{12}\sum_{p=1}^{P}\sum_{d=1}^{D}\sum_{r=1}^{R_d} z_DC_{t,p,d,r} * TrFC_DC_{p,d,r}\right) \end{aligned} \quad (3)$$

iv. CO_2 *Emission*

$$\begin{aligned}\text{EC} = &\sum_{t=1}^{T}\sum_{v=1}^{V}\sum_{d=1}^{D} e_v * d_PD_d * OZ_PD_{t,v,d} \\ &+ \sum_{t=1}^{T}\sum_{v=1}^{V}\sum_{d=1}^{D}\sum_{k=1}^{D} e_v * d_DD_{d,k} * OZ_DD_{t,v,d,k} \\ &+ \sum_{t=1}^{T}\sum_{p=1}^{P}\sum_{d=1}^{D}\sum_{r=1}^{R_d} e_{tr} * d_DC_{d,r} * z_DC_{t,p,d,r} \end{aligned} \quad (4)$$

v. *Objective Function*

$$\begin{aligned}\text{Min Z} &= \lambda_1 * (\text{MaC} + \text{TTrC} + \text{InvC}) + \lambda_2 * \text{EC} \\ &= \lambda_1 * (\text{Total Material Cost} \\ &\quad + \text{Total Transportation Cost} \\ &\quad + \text{Total Inventory Cost}) + \lambda_2 \\ &\quad * \text{total quantity of } CO_2 \text{ Emission} \\ &= \lambda_1 * \left[\left(\sum_{t=1}^{T}\sum_{p=1}^{P}\sum_{c=1}^{Comp_p}\sum_{m=1}^{Raw_{p,c}}\sum_{j=1}^{S_{p,c}}\sum_{i=1}^{S'_{p,m,c,j}} \Pr'_{p,m,c,i} \right.\right. \\ &\quad * q_Raw_{t,p,m,c,i,j} \\ &\quad \left.\left. + \sum_{t=1}^{T}\sum_{p=1}^{P}\sum_{c=1}^{Comp_p}\sum_{j=1}^{S_{p,c}} \Pr_{p,c,j} * q_Comp_{t,p,c,j}\right) \right. \\ &\quad + \left(\sum_{t=1}^{T}\sum_{p=1}^{P}\sum_{c=1}^{Comp_P} h_{p,c}\,.\,Inv_M_{t,p,c} \right. \\ &\quad + \sum_{t=1}^{T}\sum_{p=1}^{P} h_p\, Inv_C_{t,p} \\ &\quad \left. + \sum_{t=1}^{T}\sum_{p=1}^{P}\sum_{d=1}^{D} h_{p,d}\, Inv_D_{t,p,d}\right) \\ &\quad + \left(\sum_{t=1}^{T}\sum_{p=1}^{P}\sum_{v=1}^{V}\sum_{d=1}^{D} q_PD_{t,p,v,d} * TrC_PD_{p,v,d} \right. \\ &\quad \left. + \sum_{t=1}^{T}\sum_{v=1}^{V}\sum_{d=1}^{D} OZ_PD_{t,v,d} * TrFC_PD_{v,d}\right) \\ &\quad + \left(\sum_{t=1}^{T}\sum_{v=1}^{V}\sum_{d=1}^{D}\sum_{k=1}^{D1_d} q_DD_{t,p,v,d,k} \right. \\ &\quad * TrC_DD_{p,v,d,k} \\ &\quad + \sum_{t=1}^{T}\sum_{v=1}^{V}\sum_{d=1}^{D}\sum_{k=1}^{D1_d} OZ_DD_{t,v,d,k} \\ &\quad \left. * TrFC_DD_{v,d,k}\right) \\ &\quad + \left(\sum_{t=1}^{12}\sum_{p=1}^{P}\sum_{d=1}^{D}\sum_{r=1}^{R_d} q_DC_{t,p,d,r} * TrC_DC_{p,d,r}\right. \end{aligned}$$

$$+\sum_{t=1}^{12}\sum_{p=1}^{P}\sum_{d=1}^{D}\sum_{r=1}^{R_d} z_DC_{t,p,d,r} * TrFC_DC_{p,d,r}\Bigg)\Bigg] + \lambda_2 * \Bigg[\sum_{t=1}^{T}\sum_{v=1}^{V}\sum_{d=1}^{D} e_v * d_PD_d * OZ_PD_{t,v,d} + \sum_{t=1}^{T}\sum_{v=1}^{V}\sum_{d=1}^{D}\sum_{k=1}^{D} e_v * d_DD_{d,k} * OZ_DD_{t,v,d,k} + \sum_{t=1}^{T}\sum_{p=1}^{P}\sum_{d=1}^{D}\sum_{r=1}^{R_d} e_{tr} * d_DC_{d,r} * z_DC_{t,p,d,r}\Bigg] \quad (5)$$

Constraint definition

There are seven different types of constraints in our model: material constraint, supplier constraints, product constraints, transportation constraints, constraint of product receiving of distributor, the focal company constraints, and distributor constraints.

i. *Material constraint*

The constraint below express the availability of each type of component and material for production, especially inventory quantity of component c stating from period $t(Inv_M_{t-1,p,c})$ need to greater than or equal to product demand quantity at this period $\left(\sum_{v=1}^{V}\sum_{d=1}^{D} q_PD_{t,p,v,d}\right)$

$$Inv_M_{t-1,p,c} \geq \sum_{v=1}^{V}\sum_{d=1}^{D} q_PD_{t,p,v,d} \quad (6)$$

(a) Raw material suppliers

Because one of our assumptions is that all materials are domestic products, and location of candidate supplier needs to close to the focal company, the raw material supplier can supply directly to the component material. The constraint below displays the quantity of raw material ($q_Raw_{t,p,m,c,i,j}$) need to be equal to the requirement of component c of supplier j ($q_Comp_{t,p,c,j}$).

$$\sum_{i=1}^{S'_{p,m,c,j}} q_Raw_{t,p,m,c,i,j} = q_Comp_{t,p,c,j} \quad (7)$$

b) Component material suppliers

At period t, the quantity of component c that is stocked in warehouse ($Inv_M_{t-1,p,c}$) plus the quantity that supplier j supply ($q_Comp_{t,c,j}$) need to meet the production requirement ($Q\Pr o_{t,p}$). The remaining of component c at the end of period will placed in the focal company's warehouse ($Inv_M_{t,c}$).

$$\sum_{j=1}^{S_{p,c}} q_Comp_{t,p,c,j} + Inv_M_{t-1,p,c} = Q\Pr o_{t,p} + Inv_M_{t,p,c} \quad (8)$$

$$t = 1 \ldots T,\ p = 1 \ldots P,\ m = 1 \ldots Raw_{p,c},$$

$$j = 1 \ldots S_{p,c},\quad c = 1 \ldots Comp_p$$

ii. *Supplier constraint*

(a) Supplier selection constraint

a.1 At raw material stage

$$\sum_{i=1}^{S'_{p,m,c,j}} z_Raw_{t,p,m,c,i,j} \leq 1 \quad (9)$$

a.2 At component material stage

$$\sum_{j=1}^{S_{p,c}} z_Comp_{t,p,c,j} \leq 1 \quad (10)$$

(b) The relationship between supplied material quantity and the routing

a.1 At raw material stage

$$q_Raw_{t,p,m,c,i,j} \leq B * z_Raw_{t,p,m,c,i,j} \quad (11)$$

$$q_Raw_{t,p,m,c,i,j} \geq z_Raw_{t,m,c,i,j} \quad (12)$$

a.2 At component material stage

$$q_Comp_{t,p,c,j} \leq B * z_Comp_{t,p,c,j} \quad (13)$$

$$q_Comp_{t,p,c,j} \geq z_Comp_{t,p,c,j} \quad (14)$$

(c) Supplier capacity constraint

$$q_Comp_{t,p,c,j} >= Q_p * z_Comp_{t,p,c,j} \quad (15)$$

$$q_Comp_{t,p,c,j} \leq Day * \Pr od_p \quad (16)$$

$$t = 1 \ldots T,\ p = 1 \ldots P,\ m = 1 \ldots Raw_{p,c},$$

$$c = 1 \ldots Comp_p,\ i = 1 \ldots S_{p,m,c,j},\ j = 1 \ldots S_{p,c}$$

iii. *Product constraint*

(a) Satisfy demand at the focal company

$$Q \Pr o_{t,p} + Inv_C_{t-1,p} = \sum_{v=1}^{V} \sum_{d=1}^{D} q_PD_{t,p,v,d} + Inv_C_{t,p} \quad (17)$$

(b) Satisfy demand of distributors

$$\sum_{v=1}^{V} q_PD_{t,p,v,d} + \sum_{v=1}^{V} \sum_{k=1}^{D1_d} q_DD_{t,p,v,k,d} + Inv_D_{t-1,p,d} = \sum_{v=1}^{V} \sum_{l=1}^{D2_d} q_DD_{t,p,v,d,l} + \sum_{r=1}^{R_d} q_DC_{t,p,d,r} + Inv_D_{t,p,d} \quad (18)$$

(c) Satisfy customer demand

$$\sum_{d=1}^{D_r} q_DC_{t,p,d,r} = DM_{t,p,r} \quad (19)$$

(d) Production capacity

d.1 Production capacity is limited by the available material quantity

$$Q \Pr o_{t,p} \le Inv_M_{t-1,p,m} + \left\{ \left[Day - \sum_{j=1}^{S_{p,c}} (z_Comp_{t,p,c,j} * \sup_t_{p,c,j}) \right] * \Pr od_p \right\} \quad (20)$$

d.2 Production capacity is limited by the capacity of the focal company such as the number of labor days and production capacity per day, or machine constraints, etc.

$$Q \Pr o_{t,p} \le Day * \Pr od_p \quad (21)$$

(e) Due date constraint

$$temp_{t,p} = \sum_{v=1}^{V} \sum_{d=1}^{D} q_PD_{t,p,v,d} \quad (22)$$

$$\frac{(temp_{t,p} - Inv_C_{t-1,p})}{\Pr od_p} + \sum_{v=1}^{V} (t_PD_{v,d} * z_PD_{t,p,v,d}) + \sum_{v=1}^{V} \sum_{k=1}^{D1} (t_DD_{v,k,d} * z_DD_{t,p,v,k,d}) + \sum_{d=1}^{D_r} t_DC_{d,r} * z_DC_{t,p,d,r} \le Day \quad (23)$$

iv. *Transportation constraint*

(a) Only one transport mode is selected to deliver product p

a.1 From the focal company to distributor at period t

$$\sum_{v=1}^{V} z_PD_{t,p,v,d} \le 1 \quad (24)$$

a.2 From distributor k to distributor d at period t

$$\sum_{v=1}^{V} z_DD_{t,p,v,k,d} \le 1 \quad (25)$$

(b) The relationship between product quantity and connection.

b.1 From the focal company to distributor d at period t

$$q_PD_{t,p,v,d} \le B * z_PD_{t,p,v,d} \quad (26)$$

$$q_PD_{t,p,v,d} \ge z_PD_{t,p,v,d} \quad (27)$$

b.2 From distributor k to distributor d at period t

$$q_DD_{t,p,v,k,d} \le B * Z_DD_{t,p,v,k,d} \quad (28)$$

$$q_DD_{t,p,v,k,d} \ge Z_DD_{t,p,v,k,d} \quad (29)$$

(c) The relationship between product quantity and connection between distributor and customer

$$q_DC_{t,p,d,r} \le B * z_DC_{t,p,d,r} \quad (30)$$

$$q_DC_{t,p,d,r} \ge z_DC_{t,p,d,r} \quad (31)$$

(d) If more than one type of product p is shipped at the same time and the same destination and the same transport mode, it will reduce transportation cost, basically, reducing the fixed transportation cost. The following equations are employed to test this situation.

d.1 If two products used the same transportation mode from the focal company to distributor, from distributor to distributor, respectively as below.

$$\sum_{v=1}^{V} OZ_PD_{t,v,d} \le 1 \quad (32)$$

$$\sum_{v=1}^{V} OZ_DD_{t,v,d,k} \le 1 \quad (33)$$

d.2 Products are shipped at the same time and the same destination from the focal company to distributor d.

$$\sum_{p=1}^{P} q_PD_{t,p,v,d} \le B * OZ_PD_{t,v,d} \quad (34)$$

$$\sum_{p=1}^{P} q_PD_{t,p,v,d} \ge OZ_PD_{t,v,d} \quad (35)$$

d.3 Products are shipped at the same time and the same destination from Distributor d to Distributor k.

$$\sum_{p=1}^{P} q_DD_{t,p,v,d,k} \leq B * OZ_DD_{t,v,d,k} \quad (36)$$

$$\sum_{p=1}^{P} q_DD_{t,p,v,d,k} \geq OZ_DD_{t,v,d,k} \quad (37)$$

v. *Constraint of product receiving of distributor*

$$\sum_{k=1}^{D1} \sum_{v=1}^{V} z_DD_{t,p,v,k,d} + \sum_{v=1}^{V} z_PD_{t,p,v,d} \leq 1 \quad (38)$$

vi. *The focal company constraint*

$$q_PD_{t,p,v,d} \geq Q_p \quad (39)$$

vii. *Distributor constraint*

$$q_DD_{t,p,v,d,k} \geq Q_p \quad (40)$$
$$t = 1 \ldots T,\ p = 1 \ldots P,\ d = 1 \ldots D,\ k = \ldots D1,$$
$$v = 1 \ldots V,\ r = 1 \ldots R,\ c = 1 \ldots Comp_p$$

Constraint decision variable

(a) All variables that represent the flow of items are integer values
$q_Comp_{t,p,c,j}, q_Raw_{t,p,m,c,i,j}, q_PD_{t,p,v,d}$, $q_DD_{t,p,v,k,d}, q_DC_{t,p,d,r}$, and $Q\,\mathrm{Pr}\,o_{t,p}$ are integer variables

(b) All variables that represent for link status are binary values
$z_Comp_{t,p,c,j}, z_Raw_{t,p,m,c,i,j}$,
$z_PD_{t,p,v,d}, z_DD_{t,p,v,d,k}, z_DC_{t,p,d,r}$,
$OZ_PD_{t,v,d}$, and $OZ_DD_{t,v,d,k}$ are binary variables

(c) All variables that represent for inventory quantity are integer values
$Inv_M_{t,p,c}, Inv_C_{t,p}, Inv_D_{t,p,d}$ are integer variables

$$t = 1 \ldots T,\ p = 1 \ldots P,\ m = 1 \ldots Raw_{p,c};$$
$$c = 1 \ldots Comp_p;\ r = 1 \ldots R$$
$$i = 1 \ldots S'_{p,m,c,j},\ j = 1 \ldots S_{p,c},$$
$$v = 1 \ldots Vd = 1 \ldots D,\ k = 1 \ldots D1$$

The numerical values testing the applicability of this model are presented in detail in the following section.

Result

Without connections among distributors (case 1 and case 2)

The algorithm proposed in our model is applied in case 1 and case 2. After 11,002 iterations and 96 s, along with 2,470 variables and 4,888 constraints, case 1 produced an optimal solution. Meanwhile, case 2 including 2,470 variables and 5,152 constraints took 16 min and 1,640,010 iterations to produce an optimal solution.

According to the experiment results, all desired information was determined, such as the material quantity that the case company should purchase from suppliers, production quantity, product quantity and transportation mode that the focal company should used to deliver to distributors, and product quantity that distributors deliver to customers at each period t. In addition, the results also have illustrated the status of each connection. If the value is zero, it means that two nodes (supplier and the focal company, the focal company and distributor, distributor and customer) are not connected. The value of objective function is synthesized in Table 6. Figures 3 and 4 take the results of period 4 of case 1 and case 2 as an example to illustrate the supply chain configuration of the case company.

With connection among distributors (case 3 and case 4)

Now we expand the previous configuration by adding the connection between distributors. In other words, distributors that are located within the EU countries or USA can supply products to each other, as shown in Fig. 5. In this Figure, the EU countries can be linked together. For instance, distributor 1 located in France can deliver products to other distributors that are located in Germany, Italy, Sweden, and Austria, and vice versa, these distributors located in those countries can also deliver products to distributor 1 (France). Besides, USA has two distributors, one located in Los Angeles and another located in New York. Therefore, these two distributors can supply products together. However, distributor i ($i = 1, 2 \ldots d$) can only receive products from one source (the focal company or another distributor).

This expanded configuration of the case company's global supply chain is applied to our proposed model and is also divided into two sub cases as previous models. No due date constraint and linked distributors is case 3; otherwise, due date constraint, and linked distributors is case 4.

The case 3 model containing 6,844 variables and 31,876 constraints took 14,537 iterations and 5 min to produce an optimal solution. Case 4 including 6,844 variables and 32,140 constraints took 4,741,260 iterations and 20 min to produce an optimal solution.

Table 6 The value of objective function value of cases 1 and 2

Case	Material cost (USD)	Transportation cost (USD)	Inventory cost (USD)	Total cost (USD)	CO_2 emission (kg)	Optimal solution
1	3,615,885	227,965	31,380	3,875,230	20,805	2,911,623
2	3,635,775	227,365	48,724	3,911,864	18,586	2,938,545

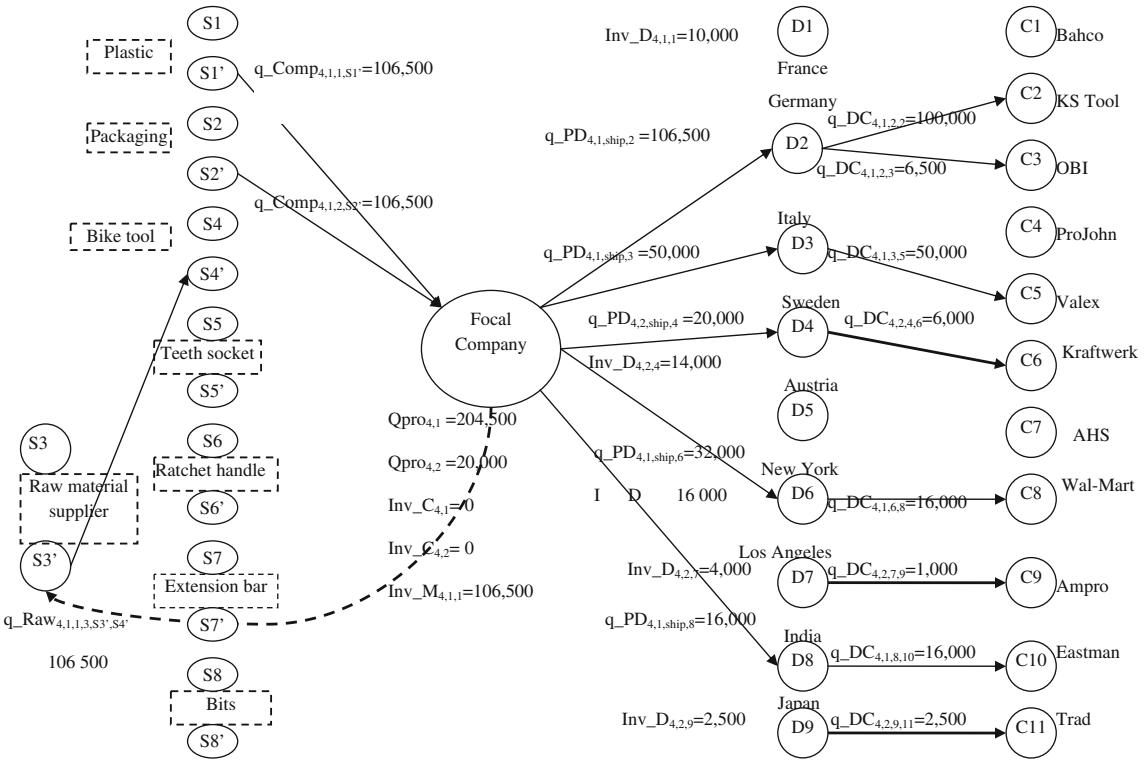


Fig. 3 Global supply chain network of the case company (case 1: no distributor connections, no due date constraint at period $t=4$)

According to the experiment results, all desired information was found the same way as in the case 1 and 2. The objective function values are shown in Table 7.

To facilitate better understanding of the results of these two cases, Figs. 6 and 7 present results of period 4 in which the most connection between nodes were made.

Global supply chain configuration results

The values of objective function are summarized in Table 8.

(a) Material cost is calculated by price of material multiplied by the purchasing material quantity. However, the price of material is offered by suppliers. So, supplier selection of the case company will affect this value (material cost).

As we can see in Table 8, the material cost of case 1 and 3 is lesser than that of case 2 and 4. The first pair (case 1 and case 3) is studied with no due date constraint situation, and the other pair (case 2 and case 4) with the due date constraint situation. At the first period when the initial inventory is zero, due date constraint forces the case company to select the suppliers with higher price and shorter lead time in order to adopt customer's lead time. The different supplier selection in the first period makes the material cost of case 2 and 4 higher than the material cost of case 1 and 3.

Furthermore, after comparing the cases without linked distributor and with distributor links, we found that when the order made by the case company is not limited by due date, the material cost is not affected by the linked distributors. Specifically, see that the material cost of case 1 and 3 is equal. Basically, the case company will select the suppliers with long lead time and cheaper price. However, when the order of the case company is limited by the due date, the material cost is affected by linked distributors. The material cost in the cases that distributors are linked is lesser than the material cost in cases in which distributors cannot be linked in two-product model.

(b) Transportation cost is calculated by fixed transportation cost (constant value) multiplied by the travel times

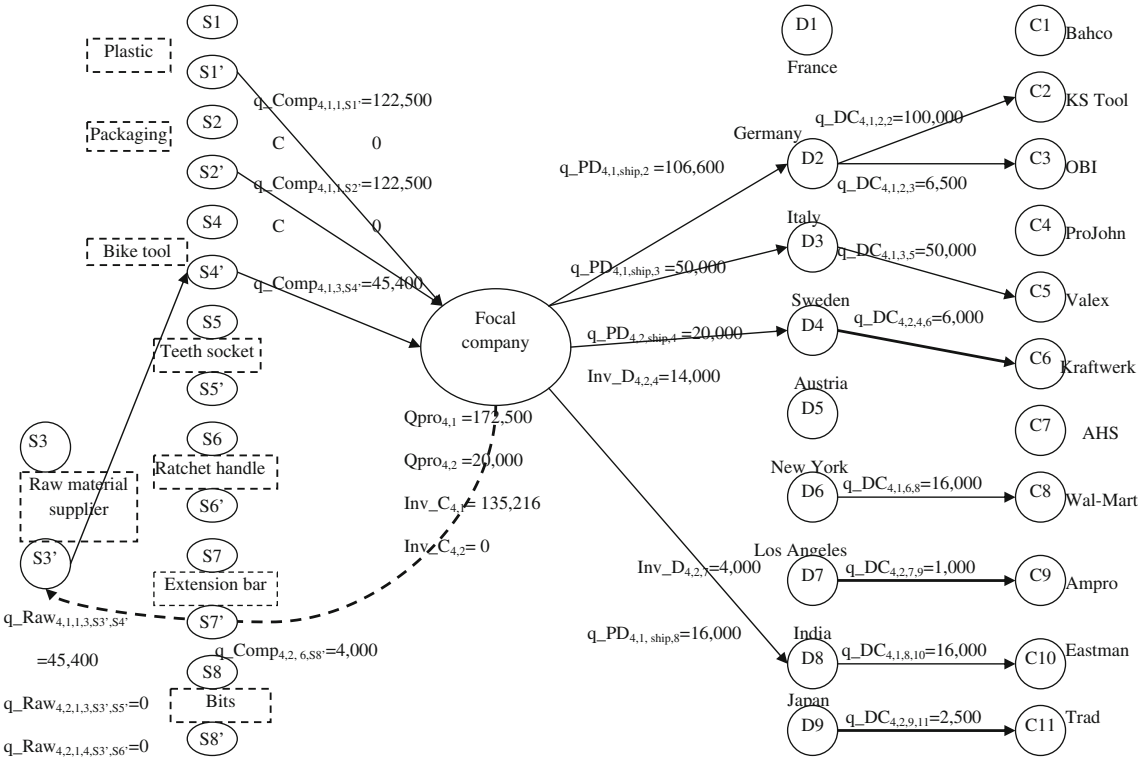


Fig. 4 Global supply chain network of the case company (case 2: without distributor connections, due date constraint at period $t = 4$). $S1$, $S1'$ are the plastic suppliers, $S2$and$S2'$ are the packaging suppliers, $S3$ and $S3'$ are the metal suppliers, $S4$ and $S4'$ are the bike tool suppliers, $S5$ and $S5'$ are the teeth socket suppliers, $S6$ and $S6'$ are the ratchet handle suppliers, $S7$ and $S7'$ are the extension bars suppliers, and $S8$ and $S8'$ are the bits suppliers. $D1$, $D2 \ldots D9$ are the distributors of the case company. $C1$, $C2 \ldots C11$ are the customers of the case company

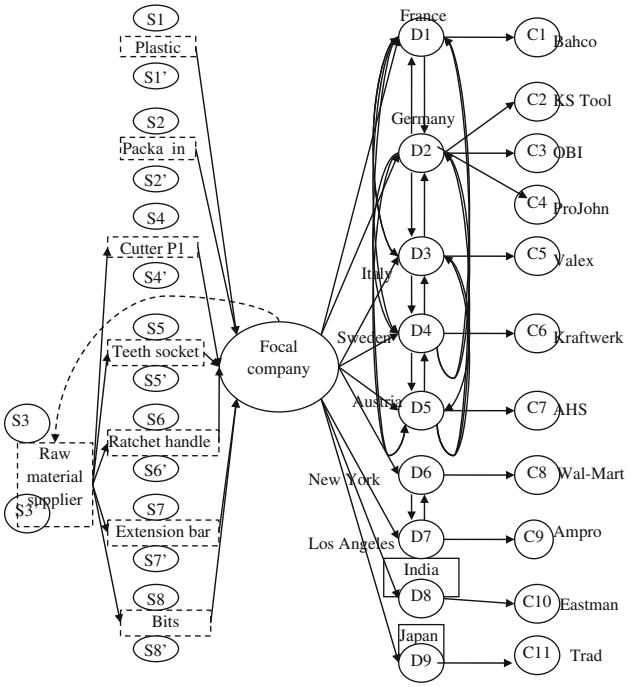


Fig. 5 The configuration of the case company's global supply chain: linked independent distributors

(or the number of shipment times) plus the transportation cost of transportation mode used multiplied by the product shipment quantity. Therefore, the transportation cost is affected by shipment times, transportation mode and product shipment quantity.

According to Table 8, the transportation cost values of case 3 and 4 are lesser than the values of case 1 and 2 since distributors are linked. In summary, the connections between distributors can also help the case company to reduce the transportation cost.

Furthermore, when we compared the no due date constraint with the due date constraint condition, we found that the transportation cost of the due date constraint is lesser than that of the no due date constraint. The case company deals with the due date constraint by increasing inventory level, therefore transportation cost is reduced.

(c) Inventory cost is calculated by inventory holding cost per unit multiplied by the quantity of products stored. This cost is influenced by number of products in stock and where the product is stored, since each location has different inventory holding cost.

Table 7 The value of objective function of cases 3 and case 4

Case	Material cost (USD)	Transportation cost (USD)	Inventory cost (USD)	Total cost (USD)	CO_2 emission (kg)	Objective function value function value
3	3,615,885	218,120	29,027	3,863,032	21,812	2,902,727
4	3,633,530	214,870	46,616	3,895,016	15,577	2,925,156

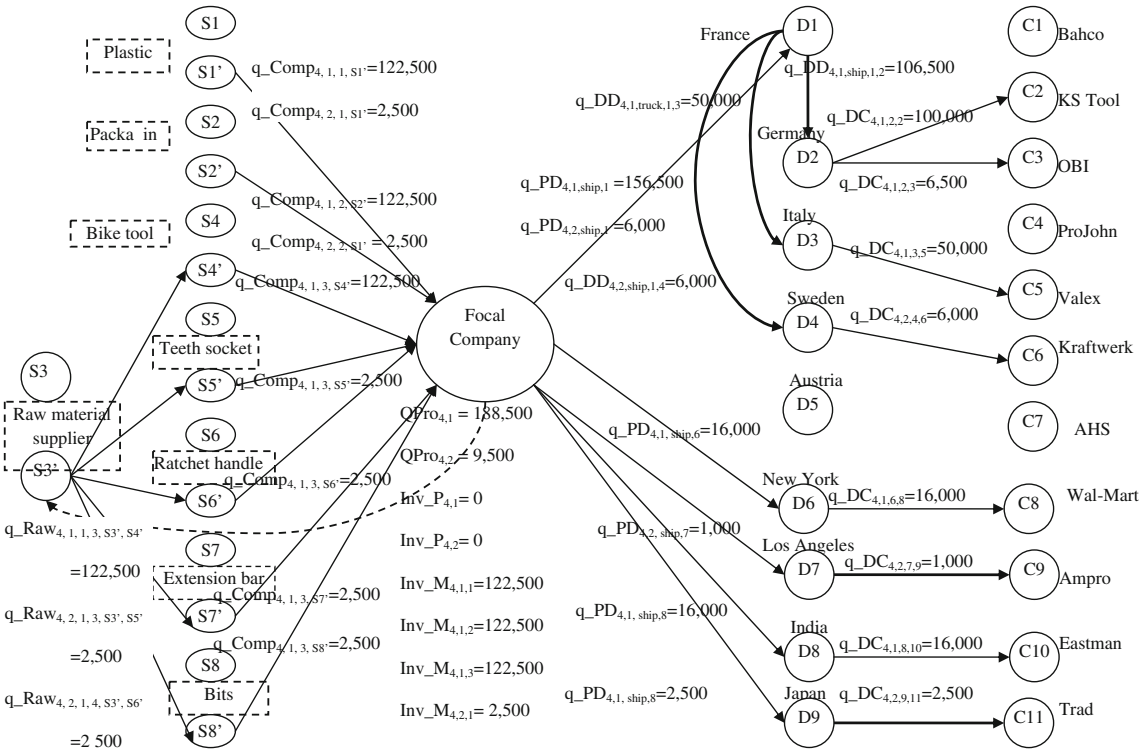


Fig. 6 Global supply chain network of the case company (case 3: distributor connections, no due date constraint at period $t = 4$)

As we can see in Table 8, the inventory cost of cases with no due date constraint are lower than the ones with the due date constraint. Specifically, the inventory cost of case 1 and 3 are lesser than that of the case 2 and 4.

We have found that having the links between distributors can reduce inventory cost. Specifically, inventory cost of case 3 that has connections between distributors is lesser than that of case 1 that has no connections between distributors. Similarly, inventory cost of case 4 that has connection between distributors is lesser than that of case 2 (no connections).

(d) Total cost is calculated by summing the three costs discussed above.

As shown in Table 8, the total cost of case 3 is lesser than the total cost of case 1, and the total cost of case 4 is lesser than the total cost of case 2. The connections between distributors help the case company to reduce the total cost.

(e) CO_2 emissions are calculated by CO_2 coefficient of transportation mode multiplied by the moving distance.

Within due date constraint, CO_2 emission of the cases that with the due date constraint is less than the CO_2 emission of the cases with no due date constraint. When the case company has the due date constraint, it increases inventory level, reduces transportation activities, thus, reduces CO_2 emission.

Within the condition of the connections of distributors, the CO_2 emission of the cases with connected distributors and without the due date constraint (case 3) is greater than the CO_2 emission of the cases that has no connections between distributors and without the due date constraint (case 1). However, the CO_2 emission of the cases with linked distributors and the due date constraint (case 4) is less than the CO_2 emission of the cases that has the connections between distributors and without the due date constraint (case 2).

(f) Objective function values are calculated by multiplying the scalar-valued weights of cost ($\lambda 1$) to total cost and by multiplying of scalar-valued weights of CO_2 emission ($\lambda 2$) to CO_2 emission value. Then they are added together.

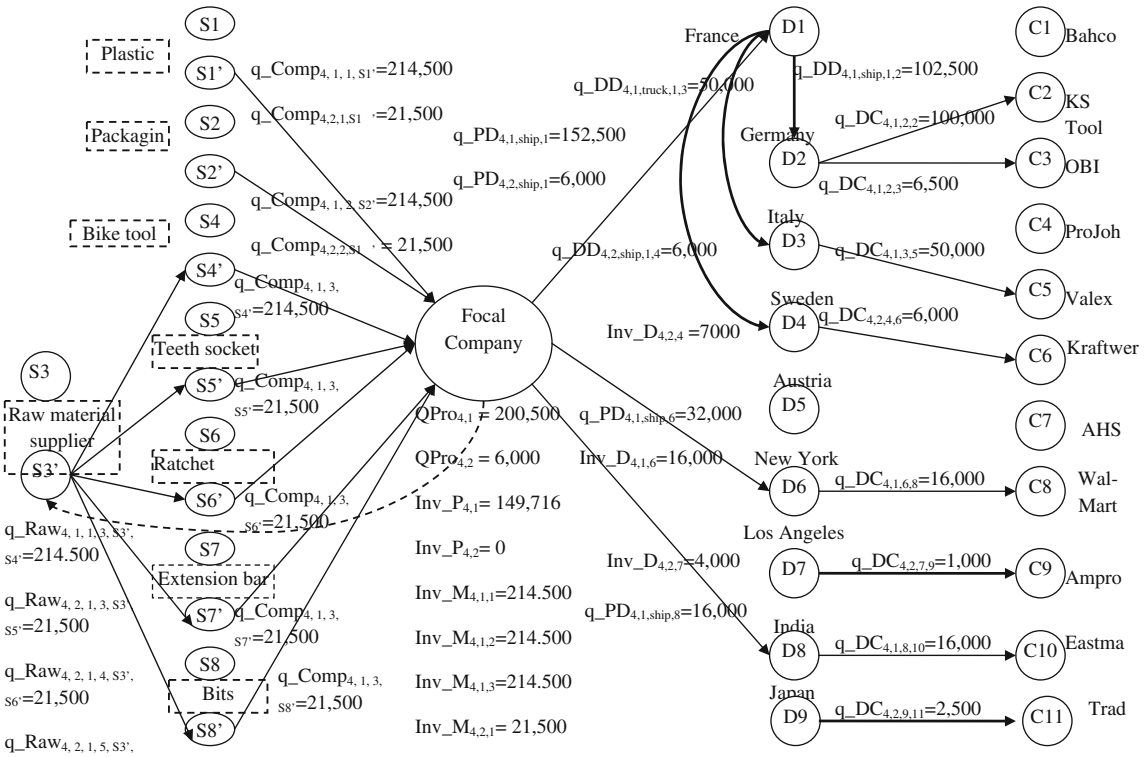


Fig. 7 Global supply chain network of the case company (case 4: distributor connections, due date constraint at period $t = 4$)

Table 8 Objective function values

Case	Material cost (USD)	Transportation cost (USD)	Inventory cost (USD)	Total cost (USD)	CO_2 emission (kg)	Objective function value
1	3,615,885	227,965	31,380	3,875,230	20,805	2,911,623
2	3,635,775	227,365	48,724	3,911,864	18,586	2,938,545
3	3,615,885	218,120	29,027	3,863,032	21,812	2,902,727
4	3,633,530	214,870	46,616	3,895,016	15,577	2,925,156

Within the due date constraint, the objective function values of case 1 and case 3 are also less than the objective function values of case 2 and 4. Within the cases of connected distributors, their objective function values are less than of the cases that with distributors who are not connected.

Generally, the connection of distributors located in different countries (especially in EU) helps the case company to reduce the total cost and objective function values. However, the CO_2 emission reaches the minimum values when the case company has both the due date constraint and connections between distributors.

In sum, Table 8 above displayed the objective function values of our four cases. The new configuration which proposes a new chance for shipping products from the focal company to the customers, the inventory level, transportation mode selection, supplier selection, and for production schedule from our experiment offers a better solution for the case company's global supply chain.

Conclusions

The following conclusions are drawn from the results obtained from four sub-models above:

(i) This research proposes an integer programming model that achieves strategic optimization of location selection, production scheduling, distribution functions, transportation modes analysis, and flow of goods. In addition, it achieves the predetermined goal of minimizing total cost and environmental impacts of CO_2 emission.

(ii) The EU scenario analysis proposes a cost-saving way for countries located outside the EU to trade with EU countries.

(iii) The transportation mode analysis and experiment results show that ship mode is still the most appealing transportation mode for the oversea shipments.

(iv) The result from the experiments of linked distributors pointed out that distributor 1 (France) is the best potential location to transfer the case company's products from Taiwan to other distributors in Europe.

(v) No matter the due date constraint exists or not, linked distributors can help the case company to save costs.

(vi) With the due date constraint, it leads a 10% difference in CO_2 between case1 and 2, while it even reaches to 28% between case3 and 4. Since due date constraint can reduce the CO_2 emission effectively, we suggest putting the due date constraint into supply chain model selection as an important factor.

(vii) The CO_2 emission can affect decisions about inventory levels and travel time of product, as well as transport mode selection. For example, we can increase inventory level to reduce the number of shipment times in order to reduce the CO_2 emission incurred by transportation activities. The transportation mode which emits less CO_2 is preferred.

(viii) Located in different countries, the product shipment between distributors using truck can reduce CO_2 emission.

(ix) This paper intended to assess the CO_2 emission and total cost in global SCM through transportation activities, inventory activities, purchasing material, etc. Moreover, the paper found out a suitable routing flow of products from suppliers to customers.

Appendix

Appendix A Parameters for multiple product model

P	The number of product types
$Comp_p$	Number of components to produce product p
$Raw_{p,c}$	Number of raw materials to produce component c of product p
$S_{p,c}$	Number of suppliers that can supply component c to produce product p
$S'_{p,m,c,j}$	Number of suppliers that can supply raw materials m to supplier j to produce component c. This component is used to produce product p
$\mathrm{Pr}_{p,c,j}$	Price of component c to produce product p that supplier j offers
$\mathrm{Pr'}_{p,m,c,i}$	Price of raw material m to produce component c of product p that supplier i offers
$\mathrm{Sup_}t_{p,c,j}$	Lead time per order that component supplier j offers to produce component c of product p
$\mathrm{Sup_}t'_{p,m,c,i}$	Lead time per an order that supplier i offers to produce raw material m. This raw material m is supplied to produce component c of product p
D	The number of distributors
$D1_d$	The number of distributors that can supply to d
$D2_d$	The number of distributors that d can supply
R	The number of customers
R_d	The number of customers that distributor d can supply
V	The number of transport modes that is used to deliver product. v = (truck, ship, air)
T	The number of periods
$DM_{t,p,r}$	Demand of product p of customer r at period t
$h_{p,c}$	Inventory holding cost of component c for producing product p at the focal company's warehouse ($US per material unit)
h_p	Inventory holding cost of product p at the focal company's warehouse ($US per product unit)
$h_{p,d}$	Inventory holding cost of product p at distributor d warehouse ($US per product unit)
$TrC_PD_{p,v,d}$	Transportation Cost per a product p unit to deliver from the focal company to distributor d using Vehicle v ($US per product)
$TrFC_PD_{v,d}$	Fixed Transportation cost per trip to deliver from the focal company to distributor d when using vehicle v ($US per trip). This cost is incurred by documental preparation fee, set-up fee, load and upload cargo fee, etc.
$TrC_DD_{p,v,d,k}$	Transportation Cost per a product p unit to deliver from distributor d to distributor k using vehicle v ($US per product unit)
$TrFC_DD_{v,d,k}$	Fixed Transportation Cost per trip to deliver from distributor d to distributor k when using vehicle v ($US per trip)
$TrC_DC_{p,d,r}$	Transportation Cost per a product p unit to deliver from distributor d to customer r ($US per product unit)
$TrFC_DC_{p,d,r}$	Fixed Transportation Cost per trip to deliver from distributor d to distributor d ($US per trip)

e_v	CO_2 emission coefficient per distance unit using vehicle v
e_{tr}	CO_2 emission coefficient per distance unit using truck
d_PD_d, $d_DD_{d,k}$, $d_DC_{d,r}$	Distances from the focal company to distributor d, from distributor d to distributor k, and from distributors d to customer r, respectively
$\Pr od_p$	Production capacity of product p that the focal company can produce per day
$t_PD_{v,d}$	Transportation time from the focal company to distributor d using vehicle v
$t_DD_{v,k,d}$	Transportation time from distributor k to distributor d using vehicle v
$t_DC_{d,r}$	Transportation time from distributor d to customer r
B	A big number value
Day	The number of days in one period
Q_p	The minimum number of Product p that can be shipped

Appendix B Variables

$q_Comp_{t,p,c,j}$	The quantity of component c that supplier j delivers to the focal company in order to produce product p at period t
$q_Raw_{t,p,m,c,i,j}$	The quantity of raw material m that supplier i delivers to supplier j in order to produce component c of product p at period t
$z_Comp_{t,p,c,j} = 1$	If supplier j delivers component c to the focal company for producing product p at period t, else = 0
$z_Raw_{t,p,m,c,i,j} = 1$	If supplier i delivers raw material m to supplier j for producing component c of product p at period t, else = 0
$q_PD_{t,p,v,d}$	The quantity of product p delivered from the focal company to distributor d using vehicle v at period t
$z_PD_{t,p,v,d} = 1$	If product p is delivered from the focal company to distributor d using vehicle v at period t, else = 0
$OZ_PD_{t,v,d} = 1$	If routing from the focal company to distributor d using vehicle v at period t is linked; else, if period t doesn't have any product p that is delivered from the focal company to distributor using vehicle v, this value = 0
$q_DD_{t,p,v,d,k}$	The quantity of product p delivered from distributor d to distributor k using vehicle v at period t
$z_DD_{t,p,v,d,k} = 1$	If product p is delivered from distributor d to distributor k using vehicle v at period t, else = 0
$OZ_DD_{t,v,d,k} = 1$	If routing from distributor d to distributor k using vehicle v at period t is linked; if period t does not have any product p that is delivered from distributor d to distributor k using vehicle v, this value = 0
$q_DC_{t,p,d,r}$	The quantity of product p delivered from distributor d to customer r at period t
$z_DC_{t,p,d,r} = 1$	If product p is delivered from distributor d to customer r at period t, otherwise = 0
$Q\Pr o_{t,p}$	The quantity of product p that the focal company produces at period t
$Inv_M_{t,p,c}$	The inventory quantity of component c to produce product p in the focal company's warehouse at period t
$Inv_C_{t,p}$	The inventory quantity of product p in the focal company's warehouse at period t
$Inv_D_{t,p,d}$	The inventory quantity of product p in warehouse of distributor d at period t
$temp_{t,p}$	Total requirement of product p that the focal company needs to supply all distributors at period t

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